

Can a Self-Driving Car's Architecture Power a Home Robot?

Introduction:

To understand vast quantities of information from sensors in real-time, advanced autonomous vehicles rely on compelling computational architectures. These designs execute complex operations, such as 3D projection, object identification, and sensor fusion, through the utilization of multi-core CPUs, vast memory structures, AI acceleration technologies, and high-bandwidth storage. As it highlights the conflicts between execution, power, and cost, the situation is extremely significant in the context of computer architecture. This study investigates the possibility of recycling an autonomous vehicle compute component for a domestic cleaning robot and recommends a modified structural layout ideal for an indoor low-energy scenario.

Background / Related Work:

The fundamentals of computer architecture maintain a significant value in utilizing storage organization, distributed computing, and specialization to maintain throughput and energy efficiency. Large storage devices, multi-level memory structures, and parallel computers are required for high-performance projects; however, they usually lead to higher power utilization, as claimed by Patterson and Hennessy (Patterson & Hennessy, 2021). Such capabilities can be justified in fully autonomous vehicles because of safety-critical delay limitations, multi-sensor integration, and real-time 3D representation.

The claimed autonomous vehicle system handles dual 4K cameras and LIDAR data with an octa-core CPU, 16 GB LPDDR5 RAM, a 1.2 TOPS AI acceleration system, and NVMe storage on an SSD. These architectures are comparable to current homogeneous systems-on-chip (SoCs) utilized in automobile platforms, such as Tesla FSD computers or NVIDIA Drive. Still, many luxurious architectural components are not required for household robots because they operate in restricted areas and typically utilize 2D SLAM, low-resolution sensors, and minimal latency requirements.

Methodology / System Design:

For a robot that cleans households, the initial vehicle design has far too many capabilities. Due to their significant power use and limited utility in an indoor atmosphere, components like dual 4K camera capability, LIDAR computing, a 1 TB NVMe SSD, and an octa-core CPU are unnecessary. Some parts remain recyclable, though, such as an efficient AI processor for human identification, an easier multi-core CPU architecture, and basic camera input capability.

A quad-core CPU, 1 MB L2 cache, 4 GB LPDDR4 RAM, an affordable AI accelerator (~0.3 TOPS), a single HD camera with thermal or ultrasonic detectors, and 64–128 GB eMMC memory make up the recommended decreased structure. Although low-power interpretation significantly boosts performance for obstacle avoidance and identification of people as compared with CPU-only performance, AI acceleration shouldn't be eliminated. Since records are only updated once a day, flash-based storage may fulfill the role of the SSD.

Component	Vehicle Module	Proposed Home Robot
CPU	Octa-core	Quad-core
L2 Cache	4 MB	1 MB
RAM	16 GB LPDDRS	4 GB LPDDR4

AI Accelerator	1.2 TOPS	~0.3 TOPS
Storage	1 TB NVMe SSD	64-128 GB eMMC
Power	~100 W peak	<10 W average

Study shows that despite occupancy meters and map buffers, memory consumption for 2D SLAM and individual monitoring usually stays below 1-2 GB. Because residential robots do not require millisecond-level response times, lowering the cache capacity and core count only marginally increases delay. It is believed that overall power usage will be reduced by about 90%, allowing for a 10-hour lifespan for the battery.

Discussion:

The recommended simplification provides sufficient performance for everyday tasks while greatly improving the effectiveness of energy and cost. With four cores, job planning across exploration, awareness, and cleanliness is still possible even with decreased similarity, reducing complexity. Lower flexibility and less ability to handle forthcoming sensors with greater resolution are the key consequences. Yet, the architecture fits extremely well with the robot's usage habits and functional restrictions.

Simplifying designs for products with scarce assets minimizes production expenses, usage of energy, and waste electronics from an environmental perspective. Considerable-performance technologies have more of an environmental impact due to their high electrical consumption and dependence on complicated conditioning and components. By enhancing technology usability while decreasing unneeded excessive engineering, reusing and modifying systems for smaller-power applications improves sustainability over time. Therefore, establishing the right equilibrium between creativity and ecological impact demands an appropriate design of architecture.

Conclusion:

Though self-driving car systems display significant computer architectural principles, this particular study reveals that they are not immediately transferable to domestic robots without modification. While keeping satisfactory performance, a more easily understood and energy-efficient structure fulfills domestic robotic demands. The suggested approach underlines the value of energy-efficient design, structural separation, and recycling with alteration as opposed to straightforward scale reduction. Innovative designs that continuously scale effectiveness in response to workload requirements may be the topic of future studies.

References:

Patterson, D. A., & Hennessy, J. L. (2021). *Computer Organization and Design: RISC-V Edition*. Morgan Kaufmann.

Sze, V., Chen, Y.-H., Yang, T.-J., & Emer, J. S. (2017). Efficient processing of deep neural networks: A tutorial and survey. *Proceedings of the IEEE*, 105(12), 2295–2329.

Quigley, M., Conley, K., Gerkey, B., Faust, J., Foote, T., Leibs, J., Wheeler, R., & Ng, A. Y. (2009). ROS: An open-source robot operating system. *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA) Workshop*.